

Exericse 1: Let $v_0, \dots, v_{m-1}$. Show that $\{v_0, \dots, v_{m-1}\}^\perp = \text{span}(v_0, \dots, v_{m-1})^\perp$.

Proof.

- Since $\{v_0, \dots, v_{m-1}\} \subseteq \text{span}(v_0, \dots, v_{m-1})$, thus $\text{span}(v_0, \dots, v_{m-1})^\perp \subseteq \{v_0, \dots, v_{m-1}\}^\perp$.
- For any $u \in \{v_0, \dots, v_{m-1}\}^\perp$ and $v \in \text{span}(v_0, \dots, v_{m-1})$, we know $v = c_0 v_0, \dots, c_{m-1} v_{m-1}$, then $\langle v, u \rangle = \langle c_0 v_0, u \rangle, \dots, \langle c_{m-1} v_{m-1}, u \rangle = 0$, thus $u \in \text{span}(v_0, \dots, v_{m-1})^\perp$

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Exericse 2: Let U a subspace of V , $u_0, \dots, u_{m-1}$ a basis of U , and $u_0, \dots, u_{m-1}, v_m, \dots, v_{m+n-1}$ a basis of V . Show that applying Gram-Schmidt Procedure to such basis of V , the result $e_0, \dots, e_{m-1}, f_m, \dots, f_{m+n-1}$. Then $e_0, \dots, e_{m-1}$ is an orthonormal basis of U and $f_m, \dots, f_{m+n-1}$ is an orthonormal basis of $U^\perp$.

Proof. Since $\text{span}(e_0, \dots, e_{m-1}) = \text{span}(u_0, \dots, u_{m-1})$, then $e_0, \dots, e_{m-1}$ is an orthonormal basis of U . For each f_k and $u \in U$, $\langle f_k, u \rangle = \langle f_k, \langle v, e_0 \rangle e_0 + \dots + \langle v, e_{m-1} \rangle e_{m-1} \rangle = 0$, since f_k is orthogonal to all e_i . Thus $f_k \in U^\perp$, therefore $f_m, \dots, f_{m+n-1}$ is a orthonormal basis of $U^\perp$ (recall that $\dim V = \dim U + \dim U^\perp$).

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Exericse 3: Let U a subspace of $\mathbf{R}^4$, defined by

$$U = \text{span}((1, 2, 3, -4), (-5, 4, 3, 2)).$$

Find a orthonormal basis of U and a orthonormal basis of $U^\perp$.

Proof. too complicated. sorry

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Exericse 4: Let $e_0, \dots, e_{n-1}$ a vector list in V , such that $\|e_k\| = 1$ for all k and

$$\|v\|^2 = |\langle v, e_0 \rangle|^2 + \dots + |\langle v, e_{n-1} \rangle|^2$$

for all $v \in V$, show that $e_0, \dots, e_{n-1}$ is an orthonormal basis of V .

Proof. Let $U = \text{span}(e_0, \dots, e_{n-1})$ and $v \in U^\perp$. Then $\|v\|^2 = |\langle v, e_0 \rangle|^2 + \dots + |\langle v, e_{n-1} \rangle|^2$ where $\langle v, e_k \rangle = 0$, thus $\|v\| = 0$, therefore $v = 0$. That means $U^\perp = \{0\}$, which means $V = U \oplus \{0\} = U$. We still need to show that $e_0, \dots, e_{n-1}$ is linear independent. For any e_k , we have $1 = \|e_k\|^2 = |\langle e_k, e_0 \rangle|^2 + \dots + |\langle e_k, e_{n-1} \rangle|^2 \geq |\langle e_k, e_k \rangle|^2 = 1^2$, thus $\langle e_k, e_i \rangle = 0$ for all j that $k \neq j$. Therefore $e_0, \dots, e_{n-1}$ is linear independent since any vector is orthogonal to each others. Hence $e_0, \dots, e_{n-1}$ is an orthonormal basis of U , thus an orthonormal basis of V .

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Exericse 5: Let V finite and U a subspace of V . Show that $P_{U^\perp} = I - P_U$.

Proof. For any $v \in V$, we can write $v = u + u^\perp$ where $u \in U$ and $u^\perp \in U^\perp$, note that the decomposition is unique, thus $P_U v = u$ and $P_{U^\perp} v = u^\perp$, and $v = P_U v + P_{U^\perp} v$. Therefore $I = P_U + P_{U^\perp}$.

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Exericse 6: Let V finite and $T \in \mathcal{L}(V, W)$. Show that $T = T P_{(\text{null } T)^\perp} = P_{\text{range } T} T$

Proof.

- We have $T P_{(\text{null } T)^\perp} v = T|_{(\text{null } T)^\perp} P_{(\text{null } T)^\perp} v$, then $(T|_{(\text{null } T)^\perp})^{-1} T v = (T|_{(\text{null } T)^\perp})^{-1} P_{\text{range } T} T v = T^\dagger T v = P_{(\text{null } T)^\perp} T v = T|_{(\text{null } T)^\perp} P_{(\text{null } T)^\perp} v = T P_{(\text{null } T)^\perp} v$.
- $T = P_{\text{range } T} T$ is trivial, since $T v \in \text{range } T$, thus $P_{\text{range } T}(T v) = T v$.

Note that we can assume $P_{\text{range } T}$ exists cause $\text{range } T$ is finite, and $\text{range } T$ is finite is by V is finite.

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Exericse 7: Let X and Y are finite subspaces of V . Show that $P_X P_Y = 0 \iff \langle x, y \rangle = 0 \forall x \in X, y \in Y$

Proof.

- ($\Rightarrow$) For any $y \in Y$, we have $P_X P_Y y = P_X y = 0$, thus $y = 0$ or $y \in X^\perp$, both case lead us to $\langle x, y \rangle = 0$ for all $x \in X$.
- ($\Leftarrow$) For any $v \in V$, $P_X(P_Y v) = 0$ since $P_Y v \in Y$, we have $P_Y v \in X^\perp$ since for any $x \in X$, we have $\langle x, P_Y v \rangle = 0$. Therefore $P_X(P_Y v) = 0$, thus $P_X P_Y = 0$. ■

Exericse 8: Let U a finite subspace of V and $v \in V$. Define $\varphi : U \rightarrow F$ by:

$$\varphi(u) = \langle u, v \rangle$$

for all $u \in U$. Apply Riesz Representation Theorem on U , we get unique $w \in U$ such that:

$$\varphi(u) = \langle u, w \rangle$$

for all $u \in U$. Show that $w = P_U v$.

Proof. We can write $v = P_U v + P_{U^\perp} v$ (by Exericse 5), then $\varphi(u) = \langle u, v \rangle = \langle u, P_U v + P_{U^\perp} v \rangle = \langle u, P_U v \rangle + \langle u, P_{U^\perp} v \rangle = \langle u, P_U v \rangle$ since $u \in U$ but $P_{U^\perp} v \in U^\perp$. Then $w = P_U v$ since w is unique. ■

Exericse 9: Let V finite. Suppose $T \in \mathcal{L}(V)$ such that $T^2 = T$ and for all $v \in \text{null } T$ and $w \in \text{range } T$, we have $v \perp w$. Show that there is a subspace U such that $T = P_U$.

Proof. By $\dim V = \dim \text{null } T + \dim \text{range } T$ and $\text{null } T \perp \text{range } T$, we have $V = \text{null } T \oplus \text{range } T$. Then for any $v \in V$, we can write $v = u + w$ where $u \in \text{null } T$ and $w \in \text{range } T$, thus $Tv = T(u + w) = Tu + Tw = Tw = P_{\text{range } T} v$. Therefore $T = P_{\text{range } T}$.

Let $v = u + w$ where $u \in \text{range } T$ and $w \in \text{null } T$ (since $V = \text{range } T \oplus \text{null } T$), and $v = u' + w'$ where $u' \in \text{range } T$ and $w' \in (\text{range } T)^\perp$ (since $V = \text{range } T \oplus (\text{range } T)^\perp$). If $v = 0$ then $u = u' = w = w' = 0$, we may suppose $v \neq 0$. Then $v - v = u + w - u' - w' = (u - u') + (w - w') = 0$, and $\langle 0, v \rangle = \langle u - u' + w - w', v \rangle = \langle u - u', v \rangle + \langle w - w', v \rangle = \langle u - u', v \rangle = 0$, which means $u - u' = 0$ since $u - u' \in \text{range } T$. Therefore $w = v - u = v - u' = w'$, and $\text{null } T = (\text{range } T)^\perp$. ■

Exericse 10: Let V finite, $T \in \mathcal{L}(V)$ such that $T^2 = T$ and $\|Tv\| \leq \|v\|$ for all $v \in V$. Show that there is a subspace U such that $T = P_U$.

Proof. I give up, I search on stackoverflow, and it seems need some limit knowledge. ■

Exericse 11: Let $T \in \mathcal{L}(V)$ and U a finite subspace of V . Show that:

$$U \text{ is invariant under } T \iff P_U T P_U = T P_U$$

Proof.

- ($\Rightarrow$) For any $v \in V$:

$$\begin{aligned} P_U T P_U v &= P_U (P_U v) \quad \text{by } U \text{ is invariant under } T \\ &= P_U v \quad \text{by } P_U \text{ is identity on } U \\ &= T P_U v \quad \text{by } U \text{ is invariant under } T \end{aligned}$$

- ($\Leftarrow$) For any $u \in U$:

$$P_U T P_U u = T P_U u$$

$$P_U T u = T u \quad \text{by } P_U \text{ is identity on } U$$

we can write $Tu = v + w$ where $v \in U$ and $w \in U^\perp$. Thus $P_U Tu = v$ and $Tu = v + w$, that means $w = 0$ thus $Tu = u \in U$. ■

Exericse 12: Let V is finite and $T \in \mathcal{L}(V)$, and U is a subspace of V . Show that:

$$U \text{ and } U^\perp \text{ are invariant under } T \iff P_U T = T P_U$$

Proof.

- By U is invariant under T , we have $P_U T P_U = T P_U$ by Exericse 11. Then

$$P_U T P_U = T P_U$$

$$P_U T (I - P_{U^\perp}) = T P_U \quad \text{by Exericse 5}$$

$$P_U T I - P_U T P_{U^\perp} = T P_U$$

We will show that $P_U T P_{U^\perp} = 0$, for any $v \in V$, we have $P_{U^\perp} v \in U^\perp$, since $U^\perp$ is invariant under T , thus $P_U T P_{U^\perp} v = P_U P_{U^\perp} v$, also, since $P_U w = 0$ for all $w \in U^\perp$, thus $P_U P_{U^\perp} v = 0$, therefore $P_U T P_{U^\perp} = 0$.

- For any $u \in U$, $P_U T u = T P_U u = T u$, thus $T u \in U$. Similarly, for any $u \in U^\perp$, $P_U T u = T P_U u = T 0 = 0$, thus $T u \in U^\perp$ since $P_U T u = 0$. ■

Exericse 13: Let $F = \mathbf{R}$ and V finite. For any $v \in V$, define φ_v a linear functorial on V by:

$$\varphi_v(u) = \langle u, v \rangle$$

for all $u \in V$.

- Show that $v \mapsto \varphi_v : V \rightarrow V'$ is injective.
- Show that $v \mapsto \varphi_v$ is isomorphism by using dimension.

Note that you should not use Riesz Representation Theorem here.

Proof.

- For any $\varphi_v = \varphi_w$ where $v, w \in V$. We have

$$\langle v - w, v - w \rangle = \langle v, v \rangle + \langle w, w \rangle - 2\langle v, w \rangle$$

Note that it is $2\langle v, w \rangle$ instead of $2 \operatorname{Re} \langle v, w \rangle$ cause $F = \mathbf{R}$. Then by $\varphi_v = \varphi_w$ we have $\langle v, v \rangle = \varphi_v(v) = \varphi_w(v) = \langle v, w \rangle$, similarly, $\langle w, w \rangle = \langle w, v \rangle = \langle v, w \rangle$, thus $\langle v - w, v - w \rangle = 0$, which means $v - w = 0$, hence $v = w$.

- $v \mapsto \varphi_v$ is surjective since $\dim V = \dim V'$ and injective, thus isomorphism. ■

Exericse 14: Let $e_0, \dots, e_{n-1}$ an orthonormal basis of V . Show that the dual basis of $e_0, \dots, e_{n-1}$ is $\varphi_{e_0}, \dots, \varphi_{e_{n-1}}$, where φ_{e_k} is given by Riesz Representation Theorem.

Proof. For any $k = 0, \dots, n-1$, let ψ_{e_k} the dual basis of e_k . We will show that $\psi_{e_k} = \varphi_{e_k}$. By Riesz Representation Theorem, we know $\psi_{e_k} = \varphi_{f_k} = \langle \cdot, f_k \rangle$ for some $f_k \in V$. By definition of ψ_{e_k} , we know $\langle e_i, f_k \rangle = 0$ for all $i \neq k$, thus f_k is a scalar multiple of e_k (Let $U = \operatorname{span}(\dots, e_{k-1}, e_{k+1}, \dots)$, then $f_k \in U^\perp$ where $U^\perp = \operatorname{span}(e_k)$). Then by $\psi_{e_k}(e_k) = \langle e_k, f_k \rangle = 1$ we know $e_k = 1 \cdot f_k$, thus $e_k = f_k$. ■

Exericse 15: In $\mathbf{R}^4$, let $U = \text{span}((1, 1, 0, 0), (1, 1, 1, 2))$. Find u such that $\|u - (1, 2, 3, 4)\|$ is minimum.

Proof. Too complicated, sorry. ■

Exericse 19: Let V finite, and $P \in \mathcal{L}(V)$ is an orthogonal projection from V to some subspace. Show that $P^\dagger = P$.

Proof. Let U a subspace of V and $P = P_U$. Then $P^\dagger = (P|_{(\text{null } P)^\perp})^{-1}P$, where $(\text{null } P)^\perp = (\text{null } P_U)^\perp = (U^\perp)^\perp = U$, and recall that P acts like identity on U , thus $P_U = (P_U)^{-1} = I$. Then $P^\dagger = IP = P$. ■

Exericse 20: Let V finite and $P \in \mathcal{L}(V, W)$. Show that:

$$\text{null } T^\dagger = (\text{range } T)^\perp \quad \text{and} \quad \text{range } T^\dagger = (\text{null } T)^\perp$$

Proof.

- $\text{null } T^\dagger = \text{null}(P_{\text{range } T}) = (\text{range } T)^\perp$. The first equation is by bijective function $((T|_{(\text{null } T)^\perp})^{-1})$ doesn't effect null space.
- $\text{range } T^\dagger = \text{range}(T|_{(\text{null } T)^\perp})^{-1} = (\text{null } T)^\perp$. The first equation is by surjective function $(P_{\text{range } T})$ doesn't effect range.

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Exericse 21: TODO

Exericse 22: Let V finite and $T \in \mathcal{L}(V, W)$. Show that:

$$TT^\dagger T = T \quad \text{and} \quad T^\dagger TT^\dagger = T^\dagger$$

These equations are obviously true when T is invertible, in that case, $T^\dagger = T^{-1}$.

Proof. Recall that $T = P_{\text{range } T}T = TP_{(\text{null } T)^\perp}$ (by Exericse 6) and $TT^\dagger = P_{\text{range } T}$ and $T^\dagger T = P_{(\text{null } T)^\perp}$. Then:

- $TT^\dagger T = P_{\text{range } T}T = T$
- $T^\dagger TT^\dagger = P_{(\text{null } T)^\perp}T^\dagger = T^\dagger$ (Recall that $\text{range } T^\dagger = (\text{null } T)^\perp$, thus the equation is also $P_{\text{range } T^\dagger}T^\dagger = T^\dagger$)

For second case, we can still use $P_{\text{range } T^\dagger}T^\dagger = T^\dagger$ even $T^\dagger : W \rightarrow V$ where W can be infinite, cause it requires that $\text{range } T^\dagger$ is finite, where $\text{range } T^\dagger$ is a subspace of V and V is finite. ■

Exericse 23: Let V and W are finite and $T \in \mathcal{L}(V, W)$. Show that:

$$(T^\dagger)^\dagger = T$$

Proof. For any $v \in V$:

$$\begin{aligned}
(T^\dagger)^\dagger &= \left(T^\dagger|_{(\text{null } T^\dagger)^\perp} \right)^{-1} P_{\text{range } T^\dagger} \\
&= \left(T^\dagger|_{((\text{range } T)^\perp)^\perp} \right)^{-1} P_{(\text{null } T)^\perp} \\
&= \left(T^\dagger|_{\text{range } T} \right)^{-1} P_{(\text{null } T)^\perp} \\
&= \left((T|_{(\text{null } T)^\perp})^{-1} \right)^{-1} P_{(\text{null } T)^\perp} \quad \text{by } P_{\text{range } T} \text{ is identity on range } T \\
&= T|_{(\text{null } T)^\perp} P_{(\text{null } T)^\perp} \\
&= T P_{(\text{null } T)^\perp} \quad \text{by range } P_{(\text{null } T)^\perp} = (\text{null } T)^\perp \\
&= T \quad \text{by Exercise 6}
\end{aligned}$$

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